

Unit 05 – Physical Layer

CONTENTS

Objectives

Introduction

5.1 Classes of transmission media

Guided Media

Unguided Media: Wireless

Switching techniques

5.2 Classification of Switching Techniques

5.3 Difference between Datagram switching and virtual circuit switching

5.4 Types of Networking Devices

Keywords

Review Questions

Self Assessment

Answers

Further Readings

Objectives

- learn what is transmission media.
- differentiate between wired and wireless media.
- learn different types of switching techniques along with their advantages and disadvantages.
- learn about the various networking devices.

Introduction

As you can see in the Figure 1, whatever medium you use to transfer data from source to destination is important. As a result, this is referred to as a transmission medium. When two people communicate with one another, a medium may be either wired or wireless. In that case, the air serves as a contact or transmitting medium. So, the transmission medium is known as air, but in this case, it is the transmission medium that is transmitting a packet from source to destination via the transmission medium, which can be wired or wireless.

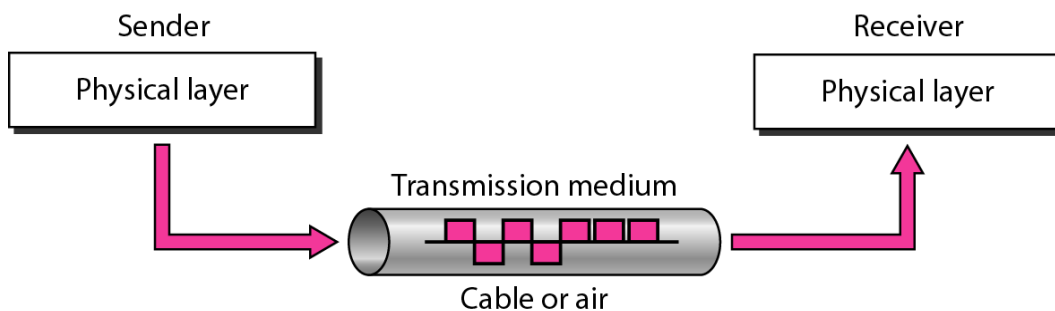


Figure 1 Transmission medium

As a result, the transmission medium is the means by which data is transmitted from one location to another. As a result, any message can be sent as data that can be translated into binary digits. As a result, the signal is encoded with these digits. The OSI model was discussed in the previous lecture. So, data is sent to a network, the network divides into packets, and the packets are formed by the network. A mail carrier truck or an aero plane could be the delivery medium for a written letter. Similarly, when 2332333

5.1 Classes of transmission media

So, there are two forms of transmission media: guided media and unguided media as shown in Figure 2. Under guided media, we have twisted pair cables, coaxial cables, and fibre optic cables.

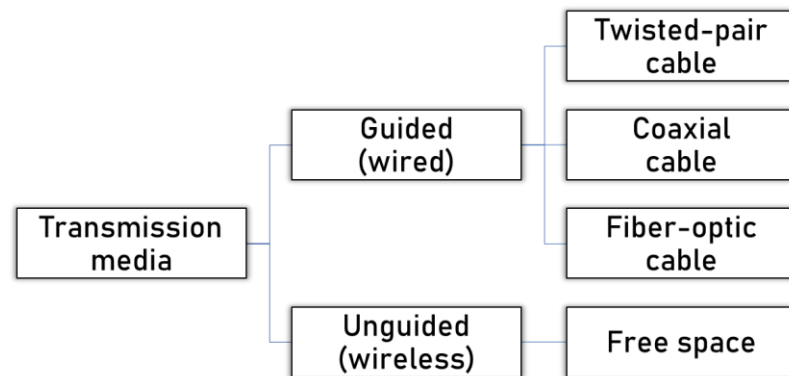


Figure 2 Classes of transmission media

Factors affecting for selecting a transmission media

So, it may be a transmission rate, cost, or ease of installation, which means that all communication media must be installed with care. The installation should be carried out by professional and experienced technicians and supervisors, and it should be resistant to the elements. Distances have an effect on the transmission medium that must be chosen, as well as the form of transmission media that must be chosen.

Guided Media

The guided media refers to a method of directing you through the process of sending data from one location to a nother through a wired link. Wired media is another term for guided media. As a result, those that are tangible or have a physical presence are those. We have a broad range of boundary transmission media. The term "boundary transmission media" refers to the use of cables or wires to link the source and destination.

Bounded transmission media are

- a. Twisted-pair cable
- b. Coaxial cable
- c. Fiber-optic cable

Twisted-pair cable

So, let's look at why we twist the wires. As you can see in the picture, if we don't twist the wires, the two wires are parallel, which means the effect of these unwanted signals isn't the same in both wires. For example, one is here and the other is here, the two parallel wires are here, some noise sources are there, which means my upper wire is closer to the noise source than the lower wire.

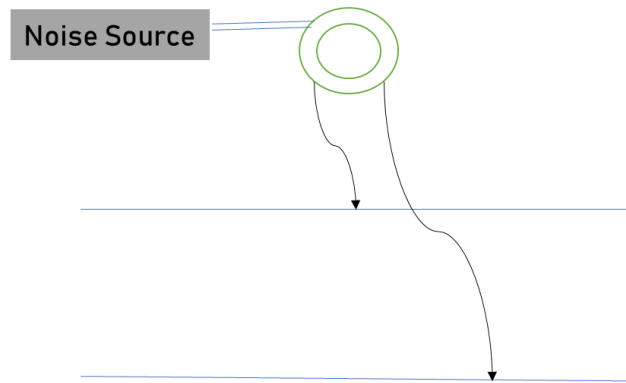


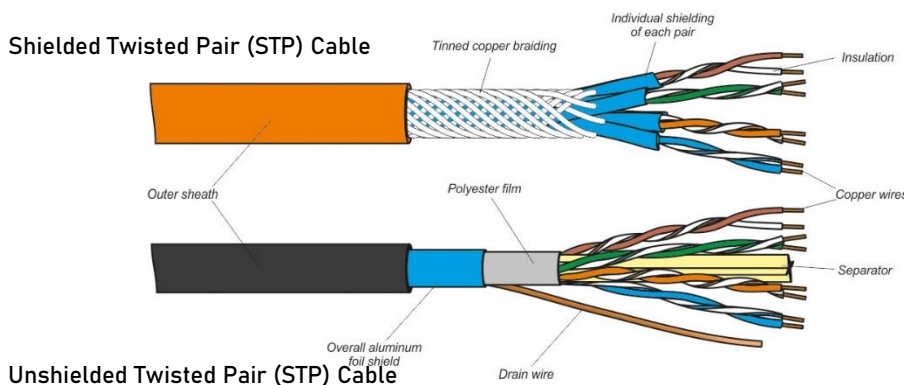
Figure 3 Effect of wire on parallel wires

that if two wires are connected in a parallel manner. As a result, my upper wire would be more affected than my other wires, owing to their different positions in relation to noise or crosstalk. As a consequence, there is a disparity at the receiver.

Similarly, we should twist the wire base to minimise crosstalk or noise because my upper wire will be close to the noise source half of the time, and my lower wire will be close to the noise source half of the time, which is why we twist the wire. So, in this case, one wire is used to transmit data and the other wire is used to receive data. As a result of the twisting, both wires are likely to be similarly influenced by external forces. Since both wires will be affected equally when we twist the wire, since half of the time this wire is closer to the noise source than half of the time this wire is closer to the noise source. As a result, the receiver that determines the difference between the two does not pick up any unnecessary signals. As a result, the undesirable signals are largely suppressed. As a result, it's obvious that the number of twists per unit of length has an effect on the cable's consistency. The higher the twist, the higher the cable's efficiency. As a result, twists often have an effect on the cable's consistency.

UTP and STP cables

Twisted pair cable comes in two varieties. Unshielded twisted pair cable is one kind, while shielded twisted pair cable is another. The most popular form of networking cable is unshielded twisted pair cable. STP is a shielded twisted pair cable that is mostly used in large organisations due to its high cost and bulkiness. It is bulkier because it has an additional metal covering.



As a result, it is bulkier, and handling this form of cable is difficult. As a result, we mostly use a UTP cable. It's not only for networking; it's also for standard telephone UTP cat 1 connections. There are various types of UTP cables available, but UTP cat five, E is the most common UTP cable, which was developed to replace the old coaxial cable, which was unable to keep up with the ever-increasing demand for faster and more reliable networks. So, we have a variety of different cable categories, and depending on which category you want to achieve, you'll need the required cable. A metal foil or braided mesh covering encases each pair of insulated conductors in shielded twisted pair cable. Both of the metal cases boost the cable's quality, and because the cable's length increases, it has an additional covering. As a result, my cable has a longer length than a UTP cable. It also has the extra covering that prevents noise or crosstalk from penetrating, as well as being bulkier and more costly. As a result, the most common twisted pair cable in communication is known as a UTP cable, which stands for unshielded twisted pair cables.

Table 1 Category of cables

CAT3:	Rarely used today, CAT3 is usually deployed in phone lines. It supports 10 <u>Mbps</u> for up to 100 meters.
CAT4	: Typically used in token ring networks, CAT4 supports 16 Mbps for up to 100 meters
CAT5:	Used in Ethernet-based LANs, CAT5 contains two twisted pairs. It supports 100 Mbps for up to 100 meters.
CAT5e:	Used in Ethernet-based LANs, CAT5e contains four twisted pairs. It supports 1 <u>Gbps</u> for 100 meters.
CAT6:	Used in Ethernet-based LANs and data center networks, CAT6 contains four tightly wound twisted pairs. It supports 1 Gbps for up to 100 meters and 10 Gbps for up to 50 meters.

Category of cables

The EIA (Electronic Industry Association) has created a set of 12 standards that divide UTP cable into seven groups. So there are categories that decide cable quality from one to seven, with one being the lowest and seven being the best, such as cat three cat 4, cat 5, cat 5 e and, cat six as shown in Table 1. So, as I previously said, cat 5e is used in Ethernet-based LANs, and we also have a cat Three that is rarely used today. Cat six is a twisted pair network that uses Ethernet. It supports one gigabit per second, up to 100 metres, and 10 Gbps for up to 50 metres.

UTP Connector

RJ 45 female and RJ 45 male connector, also known as registered jack, are the most common connectors in the UTP.

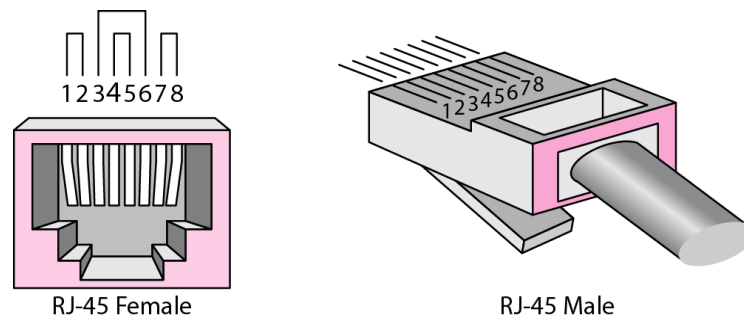


Figure 4 UTP Connector

Advantages of Twisting

It eliminates crosstalk, or the flow of information from one buyer to another. It also protects against external signal interference. It's simple, versatile, and attached, which means that if you want to extend the cable's length, you can do so easily. It's also simple to install and maintain; the steps are straightforward, and there's no need to learn any complicated formulas. It's also lightweight and inexpensive. It is available for purchase by everyone.

Disadvantages of Twisted Pair Cable

A signal with a high attenuation has been lost. Carrying a signal over a long distance without a repeater is a challenge because when the signal loses energy and becomes weaker and weaker, it becomes a

problem. So, in that case, I'll need a repeater, and the role of a repeater is to improve the signal. So, low bandwidth, maximum data range from one megabit per second to ten megabits per second, without any additional equipment repeater is also hardware and incurs costs, so this is one of the disadvantages.

5.2.2 Coaxial Cable

Since the two media are built so differently, coaxial cable can carry signals with higher frequency ranges than twisted pair cable. Coax has a solid or regular wire centre core that is surrounded by an insulated layer, as seen in the Figure 5.

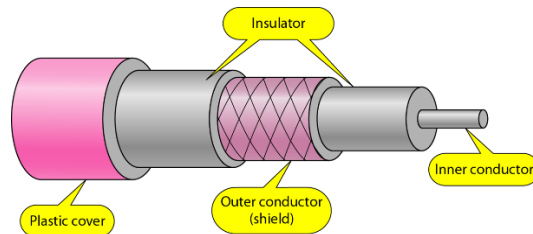


Figure 5 Coaxial Cable

As a result, cable TV connectors companies are the most popular users of coaxial cable. So, there are various types of coaxial cables, and they are classified as coaxial cables based on their radio government ratings.

Connectors

The most popular coaxial cable connectors are Bayonet Neill connectors, which are used to attach coaxial cable to the devices that require coaxial connectors. Now, as you can see in the diagram, the most popular connectors are BNC, BNC T connector, and BNC Terminator as shown in Figure 6.

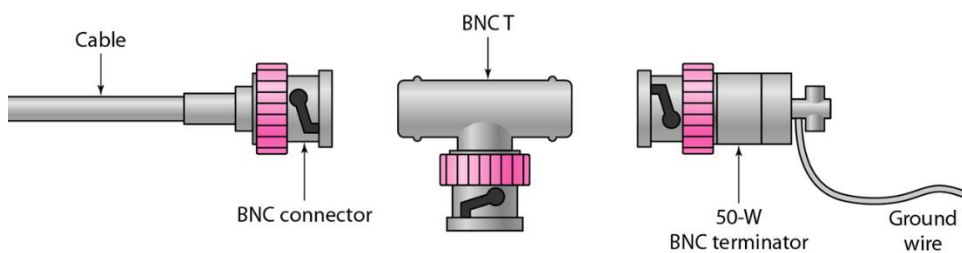


Figure 6 BNC Connector

There are three different types of connectors. As a result, BNC connectors are used to link the end of a cable to a system such as a television set. To avoid signal reflection, a BNC Terminator is used at the cable's end. While coaxial cable has a far higher bandwidth than twisted pair cable, attenuation is far higher in this case, and the signal weakens quickly, necessitating the use of repeaters on a regular basis.

Performance of coaxial cable

While coaxial cable has a far higher bandwidth than twisted pair cable, attenuation is far higher in this case, and the signal weakens quickly, necessitating the use of repeaters on a regular basis. The responsibilities of coaxial cables and how they are used in television signals. It's used to link computers in local area networks, and shorter lengths of this cable are used to connect machines to test equipment including signal generators. It is also used to transmit radio signals.

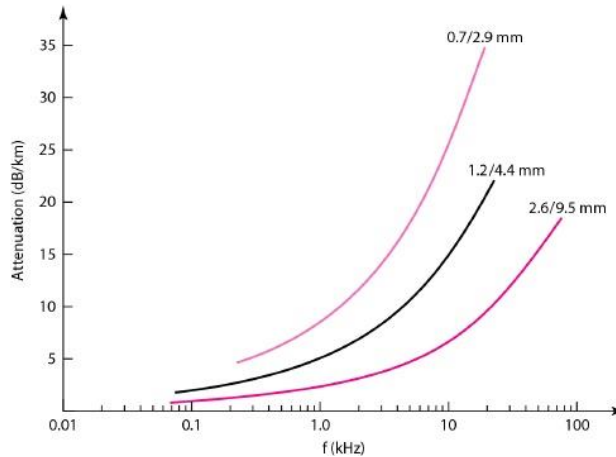


Figure 7 Performance of coaxial cable

Categories of Coaxial Cables

There are two types of coaxial cable available: thin net and thick net Figure 8. In reality, thick net is still used for backbone wiring, and twisted pair cable is an alternative to thinnet on an Ethernet network. Thicknet was the initial Ethernet wiring, but thinnet is less expensive and easier to install. In the case of thick net, the maximum length is 500 metres; in the case of thin net, the maximum length is 185 metres. Its names, thicknet and thin net, are self-evident. The term "thick net" refers to a cable that is thicker than the term "thin net," and the term "thin net" refers to a cable that is thinner. Table 1 shows the disparity between thicknet and thinnet.

Thicker	Thinnet
	
• Coax Cable RG-8 • 10Base5 • Thicknet Cable	• Coax Cable RG-58 • 10Base2 • Thinnet Cable (Cheapernet)

Category	Impedance	Use
RG-59	75	Cable TV
RG-58	50	Thin Ethernet
RG-11	50	Thick Ethernet

Figure 8 Categories of coaxial cable

Table 2 Difference between thicknet and thinnet

Thicknet	Thinnet
Thicker than thinnet	Thinner
Max length 500 meters	Max length 185 meters

Advantages of Coaxial cable

Now consider the benefits of coaxial cable transmission rate over twisted pair cable. It can be used in a mutual cable network and is mostly used for broadband transmission. Both channels are sent to the same wire at the same time. It has a high bandwidth of 400 megabits per second.

Disadvantages of Coaxial Cable

It's pricey, and it's not compatible with twisted pair cable if you want to extend the cable's duration. Since coaxial and twisted pair cables are incompatible, they cannot be linked. As a result, this is one of the most significant disadvantages of a coaxial cable.

Fibre Optic Cables

I believe it is also obvious from the name. And everyone knows that when transmitting data, the speed is extremely high, particularly when compared to twisted and coaxial cables. Fiber optic cables are also used in medical equipment. It is made up of 1000s of strong fibres. As a result, this single strand of fibre is as fine as a human hair. As a result, it transmits data in the form of light shown in Figure 9. This is why it's called fibre optic cable. So, it's made up of an inner glass core surrounded by a glass-like substance with a low refractive index, and the fibre optic cable's core cladding and protective coating.

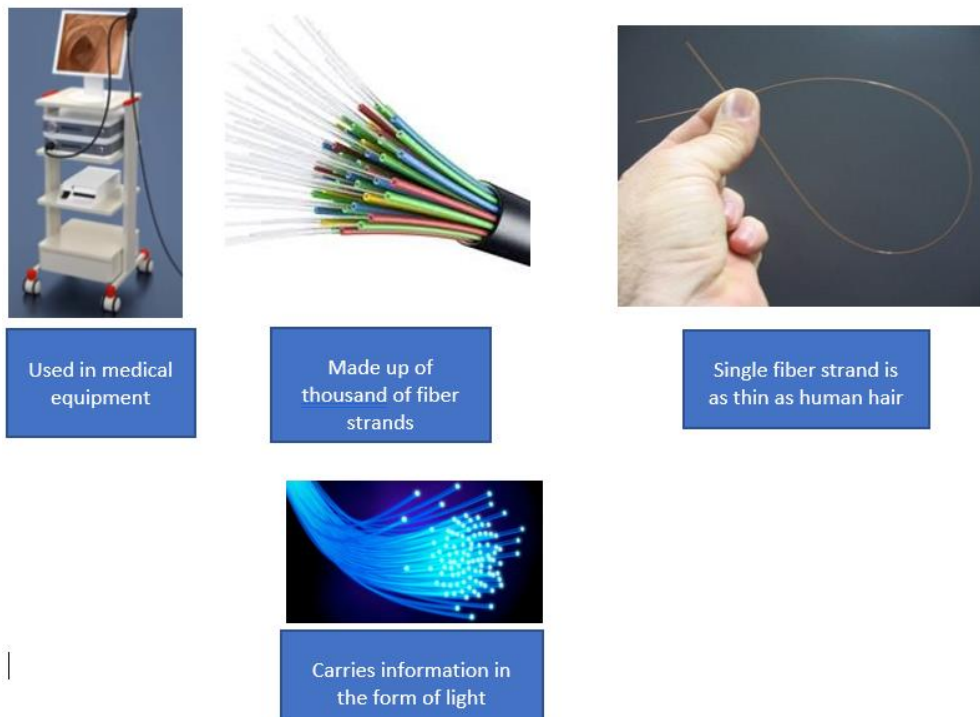


Figure 9 Fibre Optic Cables

To comprehend the workings of a fibre optic cable, we must first comprehend the fundamental behaviour of light. As light travels through a medium, such as a medium or a container, the speed of the light changes. The refractive index is determined by dividing the speed of light in a vacuum by the speed of light in a medium Figure 10. So while this shift in pace is a fascinating phenomenon, it also brings with it a slew of new words to learn, one of which is refraction.

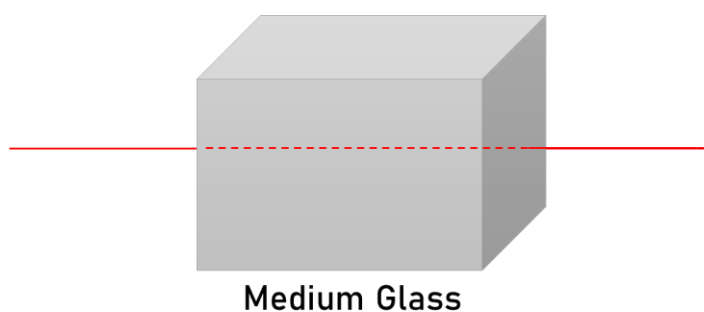


Figure 10 Medium Glass

As light travels through a prism, it bends instead of going straight, as seen in the Figure 11. When light travels from a medium with a high reflective index to one with a lower reflective index, it bends towards the interface, which is known as refraction.

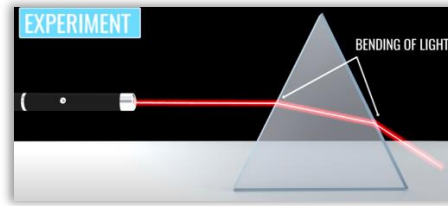


Figure 11 Refraction

Refraction is the explanation why a pencil appears bent in a glass of water, as an example. As a result, the fibre optic cable employs this basic strategy. So, as we raise the refractive index of the glass in real time, the light will bend more and more to the surface; as you increase the refractive index, the light will suddenly come to the first medium as a pure reflection; as you increase the refractive index, the light will bend more and more to the surface; as you increase the refractive index, the light will suddenly come to the first medium as a pure reflection. A complete internal reflection is what it's called. So, rather than raising the refractive index, we should increase the incident angle in this situation, resulting in a critical angle.



Figure 12 Pencil appear bent in a glass of water

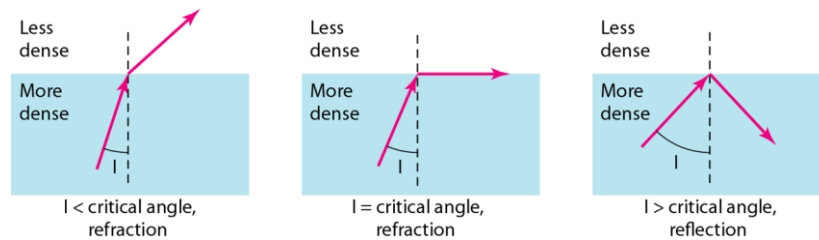


Figure 13 Bending of light ray

When the angle of incidence approaches the critical angle, the ray reflects and moves closer to the surface; as the angle approaches the critical angle, the light bends around the interface. If the angle of incidence is greater than the critical angle, the ray reflects and passes through the denser material again as seen in Figure 13.

Let's look at propagation modes now.

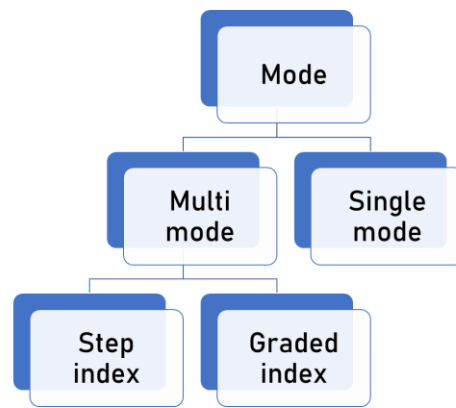


Figure 14 Propagation Modes

So, there are two modes available. One is multimode, while the other is single mode. Multimode is further subdivided into step and graded indexes. So, let's take a look at each one individually. So, what is a multimode name? It refers to the fact that multiple light beams from a single light source pass through the heart in different directions. However, the movement of these beams inside the cable is dependent on the structure; in many cases, the core density remains constant from the centre to the edges. A beam of light travels in a straight line through this constant density until it reaches the interface of the core and the cladding. At the interface, there are unexpected changes due to a lower density. As a result, the angles of a beam motion are alternated. Let's talk about this step index mode now.

In the case of step index, the suddenness of the transition means that there is an abrupt change in the signal, which leads to signal distortion as it travels through the fibre as seen in Figure 15.

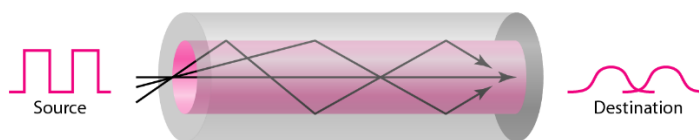


Figure 15 Multimode, Step Index

As you can see in the Figure 16, multimode graded index fibre reduces signal distortion as it travels through the cable. The term index here refers to a reflection index.

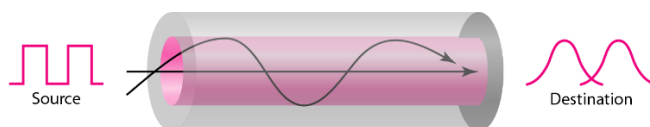


Figure 16 Multimode, Graded Index

Let's talk about single mode now. Single mode now employs a phase index fibre and a highly oriented light source to confine beams to a narrow range of angles, all of which are similar to the horizontal. As you can see Figure 17, since the diameter in single mode is so small, my light is only going in one direction, and there is only one straight line. Since the diameter is so small, it can't go up and down; instead, it can only move in one direction: up and down. As a result, single mode uses a phase index fibre and a highly oriented light source to confine a beam to a narrow range of angles all close to the horizontal.

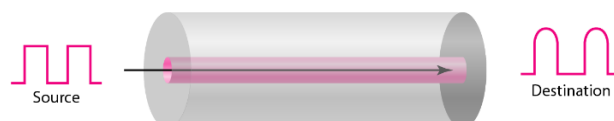


Figure 17 Single Mode

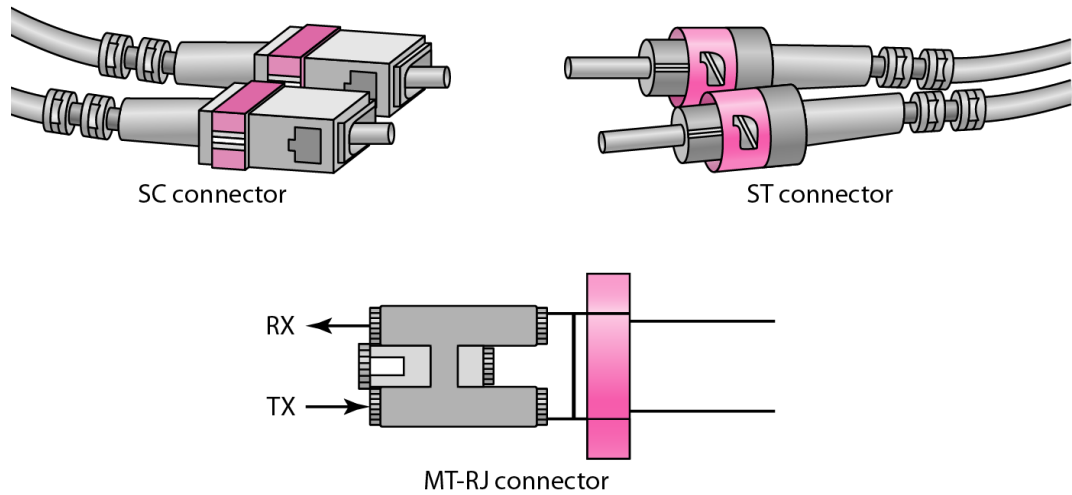


Figure 18 Fibre optic cable connectors

So, there are fibre optic cable connectors. There are SC connectors and ST connectors. These are the names of the connectors used in the five most popular scenarios.

Advantages of Fiber Optic Cable

- It is immune to electrical/mechanical interface.
- Highly suitable for harsh environment
- Secure transmission
- Used for broadband transmission.

Disadvantages

- Installation is difficult(glass is very fragile)
- Connecting two fibers is difficult.
- Connection loss
- Most expensive

Unguided Media: Wireless

Now let's talk about the unguided media that carry electromagnetic waves without the use of any physical conductor. Unguided means that we're not using cables to relay signals; instead, we're using waves that may be electromagnetic waves.

Types of unguided media: -

- Radio Waves
- Microwaves
- Infrared

Unguided media, as seen in the diagram, transport electromagnetic waves without the use of a physical conductor. As a result, signals are typically transmitted over an open space and are thus accessible to anyone with a platform capable of receiving them. As a result, unguided media will move from point A to point B in a variety of ways.

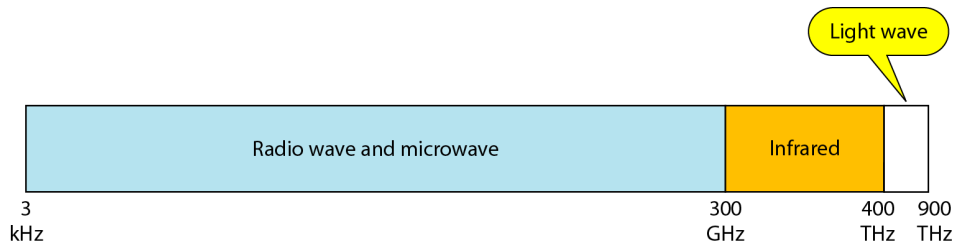


Figure 19 Electromagnetic Spectrum for Wireless Communication

Propagation Methods

Ground, sky, and line of sight propagation are seen in Figure 20. As a result, in ground propagation, the antenna flies to the lowest part of the atmosphere, hugging the ground. As a result, the lowest frequency signals emitted from the transmitting antenna obey the curvature of the earth in all directions.

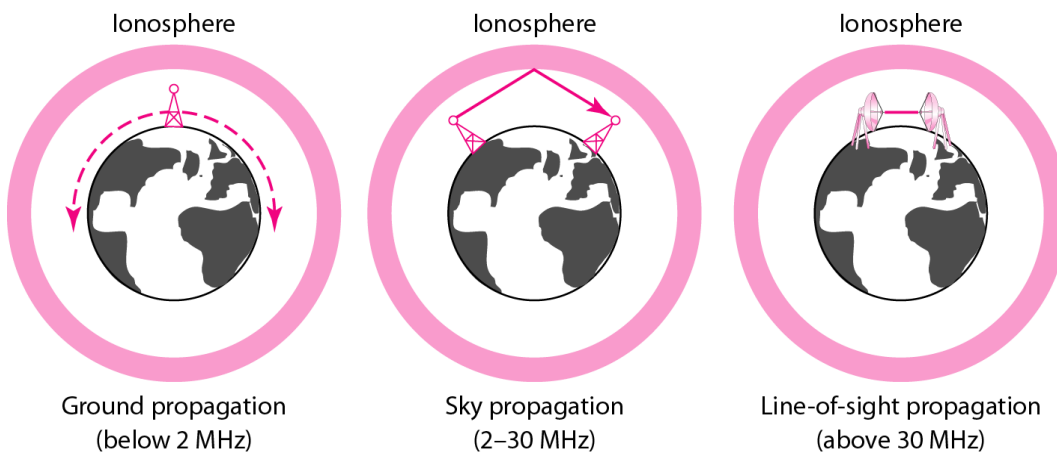


Figure 20 Propagation Methods

The amount of power in the signal determines the distance it can cover; the greater the power, the greater the distance it can cover. What about higher frequency radio waves that propagate upward through the ionosphere? They mirrored back to Earth there. As a result, this method of transmission allows for longer distances while using less fuel. As a result, very high frequency signals are transmitted in straight lines via line of sight propagation. As a result, antenna to antenna is what it means. The signal is transmitted antenna to antenna, whether they are tall apart or wherever they are.

Wireless Transmission Waves

As seen in the Figure 21 below, there are three forms of wireless communication.

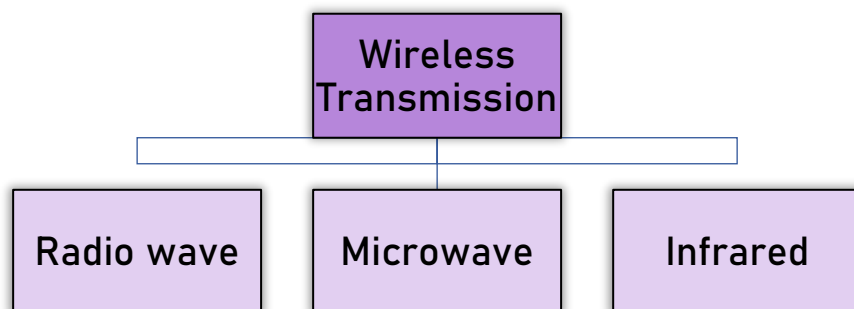


Figure 21 Types of Wireless Transmission

Radio waves

It's a method of data transmission that uses radio waves rather than copper or glass, so energy flows through the air. Radio, television, cellular phones, and other devices use radio communication in some way. Radio waves can pass through walls and even an entire structure. They can travel long or short distances depending on the frequency. Radio waves are used for multicast communications, such as radio and television, and paging systems. They can penetrate through walls. Highly regulated. Omni directional or directional antennas are used to broadcast radio waves depending upon band Figure 22. The transceiver unit, which is consisted of transmitter and receiver along with the antenna, determines the power of RF signal. Other characteristics of radio waves is that in vacuum all electromagnetic waves or radio waves travel at the same speed i.e., at the speed of light which is equal to 3×10^8 meter per seconds. In any medium this speed gets reduced and also becomes frequency dependent. In case of copper the speed of light becomes approximately two thirds of the speed of light.

- The basic features of the radio waves are that:
- They are easy to generate
- They have same velocity in vacuum
- They may traverse long distances
- They are omni directional
- They can penetrate building easily so they find extensive use in communication both indoor and outdoor
- They are frequency dependent. At low frequency they can pass through obstacles well but the power falls off sharply with distance from the source, as power is inversely proportional to cube of the distance from the source. At HF they travel in straight lines and bounce off obstacles.

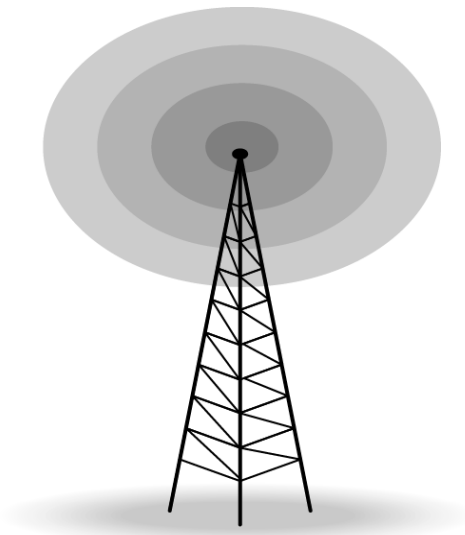


Figure 22 Omnidirectional Antenna

Advantages of Radio Wave

- Inexpensive mode of information exchange.
- No land needs to be acquired for laying cables.
- Installation and maintenance of devices is cheap.

Disadvantages of Radio Wave

- Insecure communication medium
- Prone to weather changes like rain, thunderstorms, etc.

MicroWave

Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs. Higher frequency ranges cannot penetrate walls. Use directional antennas - point to point line of sight communications. Microwave radio, a form of radio transmission that uses ultra-high frequencies, developed out of experiments with radar (radio detecting and ranging) during the period preceding World War II. There are several frequency ranges assigned to microwave systems, all of which are in the Giga Hertz (GHz) range and the wavelength in the millimeter range. This very short wavelength gives rise to the term microwave. Such high frequency signals are especially susceptible to attenuation and, therefore must be amplified or repeated after a particular distance. In order to maximize the strength of such a high frequency signal and, therefore, to increase the distance of transmission at acceptable levels, the radio beams are highly focused. The transmit antenna is centered in a concave, reflective metal dish which serves to focus the radio beam with maximum effect on the receiving antenna. The receiving antenna, similarly, is centered in a concave metal dish, which serves to collect the maximum amount of incoming signal.

Unidirectional Antennas

Unidirectional antennas, which send signals in only one direction, are needed for microwaves. For microwave communications, two types of antennas are used: the parabolic dish and the horn antenna, as shown in Figure 23. The geometry of a parabola is used to build a parabolic dish antenna: Any line parallel to the line of symmetry (line of sight) reflects off the curve at different angles, resulting in a point called the focus where all the lines converge. The parabolic dish acts as a funnel, collecting a variety of waves and funnelling them to a single point.

A horn directed at the dish is used to broadcast outgoing transmissions. The microwaves collide with the dish and deflect outward, reversing the receipt route. A horn antenna resembles a massive scoop. Outgoing signals are broadcast up a stem (which resembles a handle) and deflected outward by the curved head in a series of narrow parallel beams. The scooped form of the horn collects received signals, analogous to a parabolic dish, and deflects them down into the stem. Microwaves are very useful when unicast (one-to-one) communication is needed between the sender and the receiver due to their unidirectional properties. They're used in mobile phones.

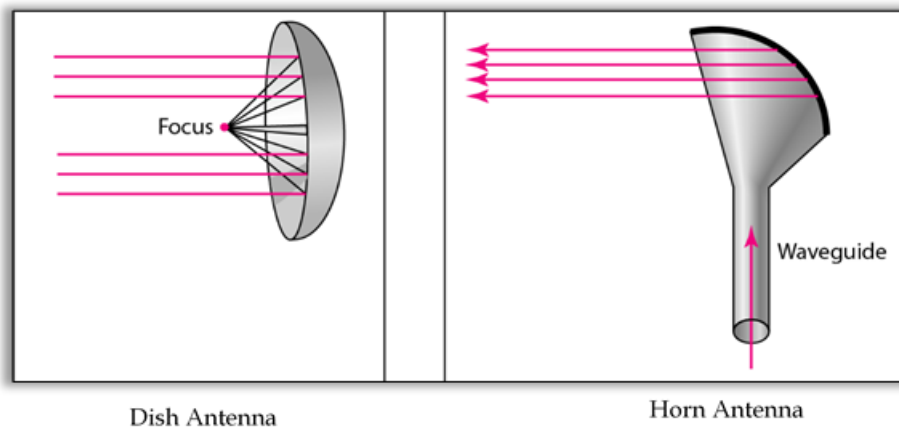


Figure 23 Unidirectional Antennas

Drawbacks

- They cannot pass through microwave buildings.
- Bad weather influences signal transmission.
- They are frequency dependent.

Infrared Signals

For short-range communication, infrared waves with frequencies ranging from 300 GHz to 400 THz (wavelengths ranging from 1 mm to 770 nm) may be used. Because of their high frequency, infrared waves are unable to penetrate walls. This beneficial feature avoids interference between systems; a short-range communication device in one room will not be affected by a system in the next room. When

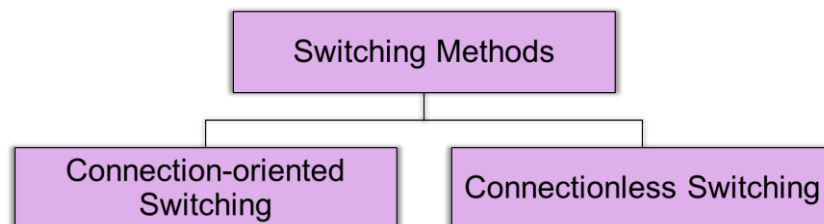
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.
- Infrared communication is unreliable outside the building because the sun rays will interfere with the infrared waves.
- Infrared waves cannot penetrate the walls.

Switching techniques

Switching Methods

So, switching approaches can be link based or connectionless.

- Connection oriented switching means that you must reserve resources before sending data.
- Connectionless switching means that you do not need to reserve resources and can begin sending data as soon as it becomes available.



5.2 Classification of Switching Techniques

Circuit Switching

LOVELY PROFESSIONAL UNIVERSITY

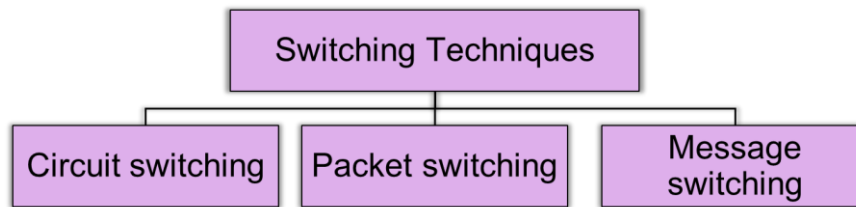


Figure 25 Switching Techniques

When you go to a restaurant and notice that the tables are already reserved, it is written on the wall. What does this mean? It means that no one else can use that table at that time because it has been reserved for a certain person, and only that person can use it. So that is what the word "dedicated" means. If the link is made, a dedicated route will remain in place until the teardown process. Circuit switching in a network functions in the same manner as it does in telephone networks. Circuit switching can be permanent or temporary. When a user tries to transmit data, voice, or video and a request, a signal is transmitted back to the receiver.

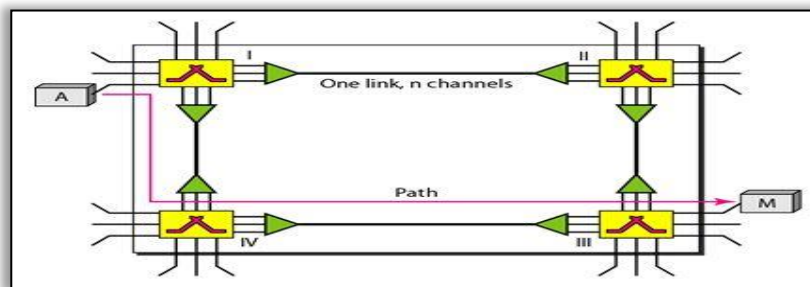


Figure 26 A trivial Circuit Switched network

The receiver then sends back an acknowledgment to ensure that the dedicated route is accessible. In public telephone networks, the data circuit switch is used after receiving the acknowledgment dedicated route transfers. It is the best example of voice communication transfer. As illustrated in the Figure 26. Each link is divided into three channels using the FdM, or TDM, and each link has three connections that operate in three phases.

Applications that use circuit switching may have to go through three phases:-

- Establishing a circuit.
- Transferring the data.
- Disconnecting the circuit

What do the terms "**establishing a circuit**" and "establishing a process" mean? Creating the circuit entails create a phase Set up a phase means that if you want to submit data, you must first reserve the resources. Channels and turn buffers are examples of resources. switch processing time switch input outputs that must be committed during the data transmission. As you can see in Figure 26, A wishes to give information to m. So, first, A will notify switch four that a dedicated path exists between switch one and switch four, then switch four will create a dedicated path between switch four and switch three, and switch four will inform node m of A's purpose, and node m will agree that it is ready to receive the data.

The **data transfer** process will only function after that. Data transfer refers to a continuous flow of data from a source station to a destination station; continuous flow implies that the data will not be split into packets because my resources have been allocated. My data does not need to be divided into packets. As a result, data will flow continuously. And the first is the setup step, in which we reserve resources such as CPU bandwidth, CPU buffer, switch processing time, and switch input output ports, which are the resources I need. Then, once the acknowledgment is received, the source station will begin sending the data, which will be sent in a continuous flow from the source station to the destinations.

Disconnecting the process is the third choice. After the data has been successfully transferred, the circuit will be disconnected.

Important points

- circuit switching, are designed for the voice applications.
- The best suitable example of circuit switching is telephone.
- The connection is established using Dedicated path.
- Circuit switching takes place at the physical layer.

Efficiency and Delay

Circuit switching has a lower performance than the other two forms of networks because it requires you to reserve resources. So keep in mind that whenever I mention reserving resources, I'm referring to a cost. To send data, I must first create a link. The total delay is due to the time required to create the link, move the connection, and disconnect the circuit; however, the delay in this type of network is very negligible because there is no waiting time because there is no connection; I have already formed a connection, so there will be no delay; the total delay is due to the time needed to create the connection, transfer the connection, and disconnect the circuit; however, the total delay is due to the time needed to create the connection, transfer the connection, and disconnect the circuit.. Otherwise, there is no delay.

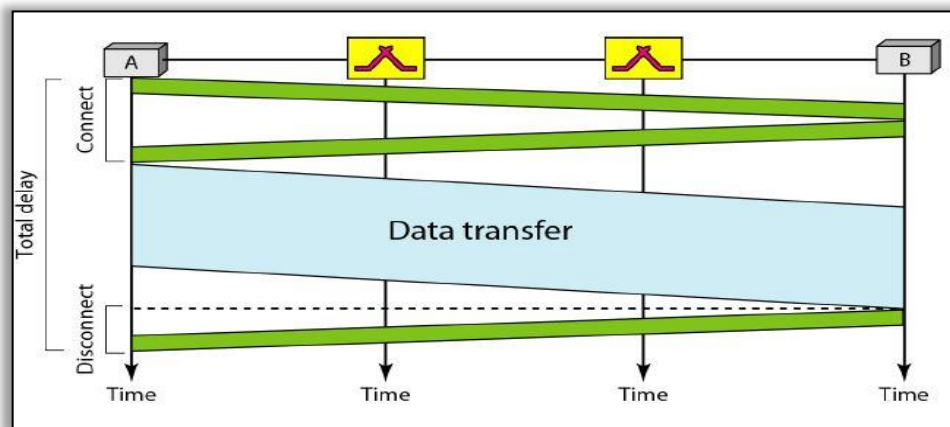


Figure 27 Delay in Circuit Switched Network

And I'm assuming that when you go to the restaurant, you won't have to wait for a table to become available because you've already reserved one. As a result, the time delay would be negligible. Only when linking, transferring, and disconnecting the circuit is there some time for delay.

As you can see in the Figure 27, the overall delay in the circuit switch network is due to the time required to establish the link, transfer data, and disconnect the circuit, so the delay in this time is very small.

Advantages of Circuit Switching

The benefits of circuit switching include the creation of a dedicated channel that no one else can use. And if I talk about dedicated, that means my packet loss will be very low, and if there is a dedicated channel, overhead will be very low because everyone is on the same path. If I want to get to a specific station and everyone is taking the same path. I'm also following the same paths because I don't have to think about which way to go, whether left or right, because I'm simply following the same direction. Improves data transfer rate significantly because there is no waiting period and no delay. As a result, the data transfer rate is strong here, which reduces data loss. If you want to go to a wedding but don't know where to go. You're just following path one after another, and some dedicated link is created for you, and you're just following the route, one after another, so data loss will be extremely rare here, and my data will not be lost. It improves delay in the data flow.

Disadvantages of circuit switching

- The waiting time lasts long, and there is no data transfer.
- Each connection has a dedicated path, and this gets expensive.
- When connected systems do not use the channel, it is kept idle.

Message Switching

As message switching is used, the entire message is treated as a data unit, and each node receives the entire message and buffers it before forwarding it to the next node. A message switching shown in Figure 28.

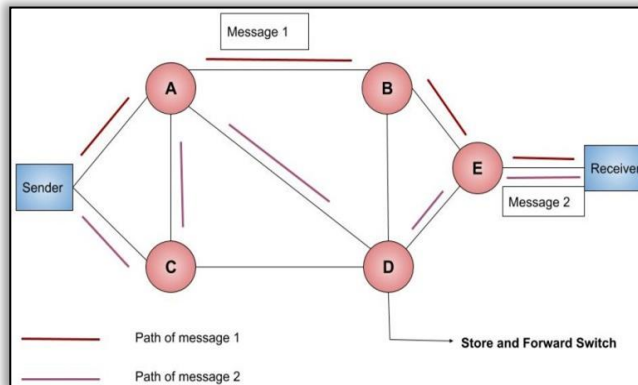


Figure 28 Message Switching

This form of network is known as a store and forward network since no dedicated route is established between the sender and the receiver. We are not allocating any resources between the sender and the recipient, implying that there is no dedicated route. It's a combination of dropping store and forward strategies.

Advantages of Message Switching

- Stores the message when the next node is not available
- Reduces traffic congestion.
- Data channels are shared by network devices.
- Manages traffic efficiently by assigning priorities.

Drawbacks of Message Switching

- Transit path needs enough storage.
- Message switching is very slow.
- Message switching was not a solution for streaming media and real-time applications.

Packet Switching

Datagram and virtual circuit packet switching are the two methods of packet switching. Let's look at what packet switching is and why it's called that because the data is split into packets. The packet switching technique is a switching method that sends the message all at once. However, it is broken down into smaller bits and sent separately. Messages are split into smaller pieces known as packets, with each packet having its own sequence number, so that we can place them in order when we receive them. The length of the packet may be set or variable; the switching information is contained in the header of each packet; the type of information header contains as shown in Figure 29. The source and destination nodes, as well as the intermediate number, are listed in the header. The sequence number is the intermediate number since, in this case, each packet is taking a different path to reach its destination. As a result, when the packets arrive, we must sort them into orders. So, who's going to assist in putting the packet together? That is your sequence number, and it is with the aid of that sequence number that we are placing the packets in the correct order. The message will be sent again if the packet is lost or corrupted.

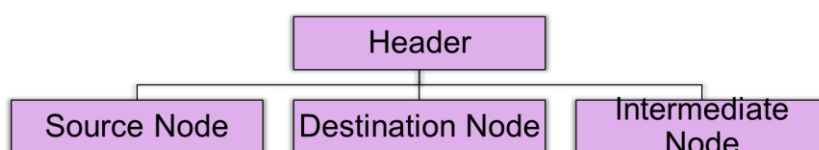


Figure 29 Details of Header

Modes of Packet Switching

So, what are the different types of packet switching? Packet switching may be connection-oriented or connectionless. In a connectionless network, packets can pass through the network using the shortest route possible, with all packets falling on separate paths to the destination being reassembled at the receiving end.

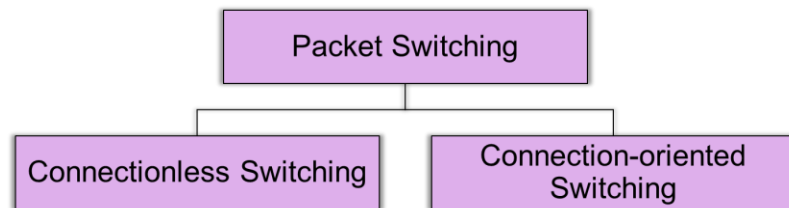


Figure 30 Modes of Packet Switching

Since this is the source and this is the destination, if I try to send data from source to destination, it will divide the data into little small packets. So that's why we'll reassemble the packet only if you're segmenting it. After you've divided the package, it's time to put it together. We'll reassemble the packets and reassemble the message when we receive them. If the correct order is met, an acknowledgment message indicating that the packet was successfully received is also sent.

Two Basic Forms of Packet Switching

There are two ways to treat packets. It may be a virtual circuit or a datagram switching as shown in Figure 31.

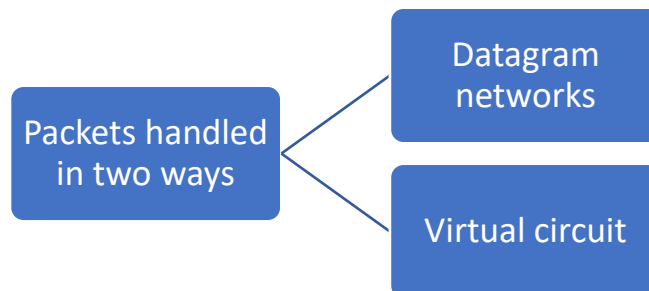


Figure 31 Forms of Packet Switching

But, first and foremost, let's talk about datagram networks.

Datagram network

It is a packet switching technology in which a packet is referred to as a datagram, hence the name datagram switching. Now, each packet contains destination information, which the switch uses to forward the packet to the next destination. Packets can now follow any practical path. So, in a datagram switching technique, the path is not set, i.e., it is not committed to the fact that you will only use this path to get to your destination. On the cloud, a variety of paths would be available. As I previously said, the modes of packet switching are connectionless and connection based, respectively, and datagram switching is a connection less switching. As a result, datagram switching, also known as connectionless switching, involves intermediate nodes making routing decisions in order to forward packets. As a result, datagram networks are also known as connectionless networks. What does connection less mean? It means that the switch does not store information about the connection state. As seen in the Figure 32 below. A is sending a packet of data, which is divided into packets, each of which has a sequence number and contains information about the source, destination, and sequence number.

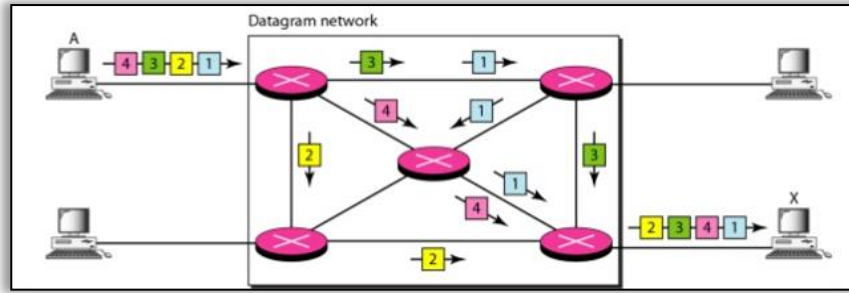


Figure 32 Datagram Network

So 1234, and as you can see here, each packet takes a different path to reach its destination, which is x. As a result, when they arrive at their destination, they are not in order 1432. This is the order, so we can place our previous message in our order by looking at their sequence number, which we can organise in the order routing table. Each switch has a routing table that is based on the destination address. Routing tables are dynamic and are modified on a regular basis if no new nodes are added to the network. Routing tables are dynamic and are updated on a regular basis if no new nodes are added to the network. As a result, the destination address now assists in getting packets to their intended destination by forwarding them to the next destination. So, in a datagram network, the destination address in the header of a packet stays the same during the entire journey of the packet performance of datagram networks is unquestionably better than a circuit switch network.

And I want to submit a packet every time. I don't need to set up a link or a phase, and I don't need to reserve resources because it costs a lot of money; resources are only allocated when there are packets to be transferred, which means resources are only reserved on demand. If you have a packet to send during datagram switching. At that time, if only you will submit the packet, which ensures that only resources will be allocated at that time. In the case of datagram switching, if you want to send a packet, resources will be allocated only on request. Since resources are not allocated in advance, there is a greater delay in a datagram network than in a virtual circuit network if you have a packet to send delay.

Virtual Circuit Networks

Since it is a combination of datagram and circuit switching, it is link driven. It is expensive to implement because if you want to transmit data, a global packet is transmitted to the network. Resources are allocated for time periods, and it is a highly efficient medium of transmission. It will also set aside all resources so that the packets can be sent to their destinations. Only if you have a package to send will the global packet be sent, which will reserve the resources as if you needed them. If you want to send a packet, you'll also need CPU bandwidth and a buffer to hold the packet for a period of time. As a result, in the case of a virtual circuit, a pre-planned route is formed prior to the message being sent. As seen in the Figure 33 below, call request and a call Accept packets are used to create the sender-receiver link. Paths are reserved in this case for the length of the logical relation. If I want to end the link, a simple request will be sent to end the connection request. So call request call accept indicates that a link has been established; if you have data to send, it will send the data transfer, and an acknowledgment will be sent indicating that my data has been received successfully; and finally, we can terminate the connection by sending a simple request.

Since a global packet is sent in this case as well, packets are sent to the recipient in the same order as they were sent by the sender. But that's how the money are set aside. As a result, every packet takes the same route to its destination. A virtual circuit is a secure network link. In each packet, there is no need for overhead.

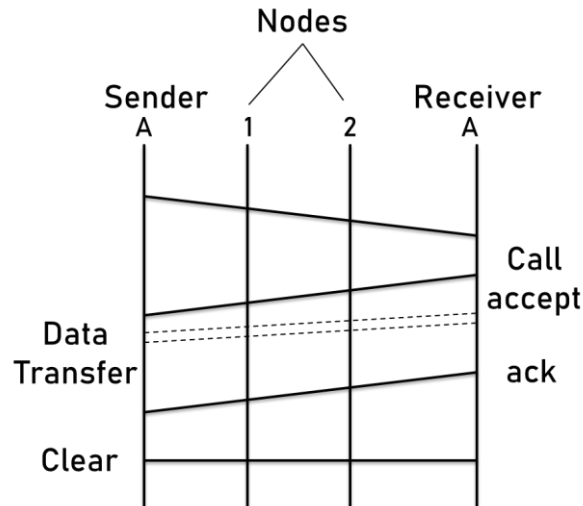


Figure 33 Working of virtual circuit

Advantages of Virtual circuit

- Packets are delivered to the receiver in the same order sent by the sender.
- Virtual circuit is a reliable network circuit.
- There is no need for overhead in each packet.
- Single global packet overhead is used in virtual circuit.

Disadvantages of Virtual Circuit

- Virtual circuit is costly to implement.
- It provides only connection-oriented service.
- Always a new connection set up is required for transmission.

5.3 Difference between Datagram switching and virtual circuit switching

The following Figure 34 illustrates the distinction between datagram switching and virtual circuit switching.

Datagram switching	Virtual circuit switching
Connectionless	Connection-oriented
No reservations	Reservations
Out of order	Same order
High overhead	Less overhead
Packet loss ↑	Packet Lost ↓
Cost ↓	Cost ↑
Delay ↑	Delay ↓

Figure 34 Datagram switching vs Virtual circuit switching

Networking Devices

As hardware devices that are used to link computers, printers, fax machines, and other electronic devices, you will be able to learn about various networking devices and understand the functioning and functionality of various networking devices. These devices send data over the same or a different network in a fast, safe, and accurate manner. As a result, we must investigate all of the devices that assist in the transfer of data from source to destination. Those are what we refer to as networking devices.

Use of Networking Devices

Figure 35 depicts the number of networking devices in operation.

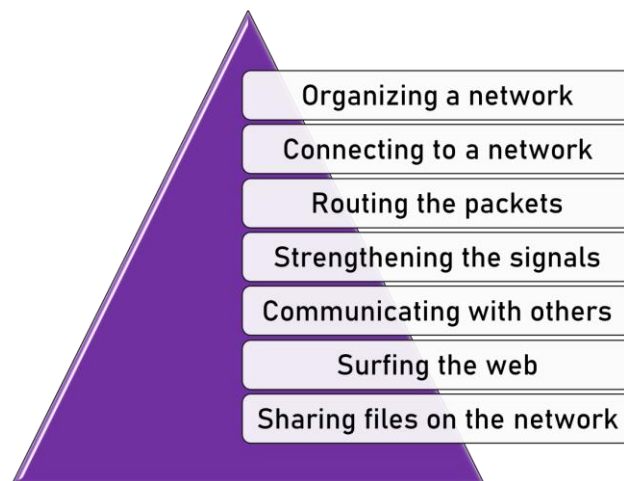


Figure 35 Use of Networking Devices

So, networking devices are the devices that are used for organizing a network, linking our network, routing the package, and improving the signals because my signal loses strength when it travels from source to destination.

5.4 Types of Networking Devices

Types of networking devices are shown in Figure 36.

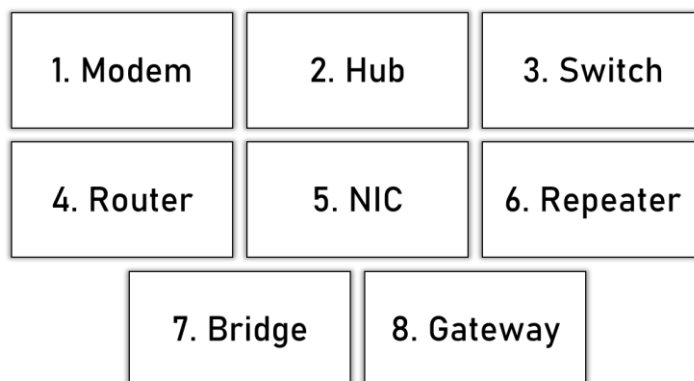


Figure 36 Types of Networking Devices

Modem

Modem is short for "Modulator-Demodulator. An analog signal is converted to a digital signal, and digital signals are converted to analog signals. The modulator is used to transform digital signals to analog signals. The demodulator modem is used to communicate over telephone lines when an analogue signal is converted to a digital signal as seen in Figure 37.

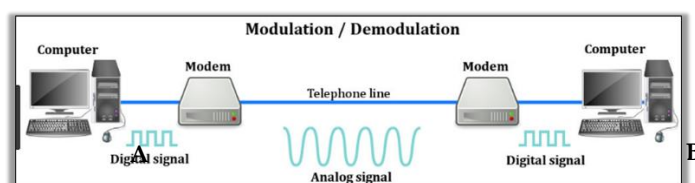


Figure 37 MODEM

How fast the modem can transfer and receive the data?

- At Slow Speeds, modems are measured in baud rate.
- At Higher Speeds, modems are measured in bits per second.
- Higher the speed, the faster you can send and receive data over the network.

Working of MODEM

So, as you can see in the diagram, if A wants to send data to B, here's an example. However, we must first use the modem before transmitting the signals to the telephone lines. So, a modem can transform digital data from a device to an analogue signal so that it can be sent over a telephone line. Then there's the fact that those cables can only accept analogue signals. Finally, we want to give a signal to machine B, which will only accept digital data. So, once again, we must use a modem to transform this analogue signal into a digital signal, which is known as a demodulator.

Which Layer Modem Works in OSI Model?

Modem works at Physical Layer of the OSI model.

HUB

The hub is a networking system that connects different types of cables to centralized network traffic via a single connecting point hub, which also serves as a central controller for all connected devices as a master controller. Hub serves as a master controller. It actually goes to whatever communication is taking place, and it will go through the hub, as seen in this example. The hub now serves as a central controller for all devices that want to connect.

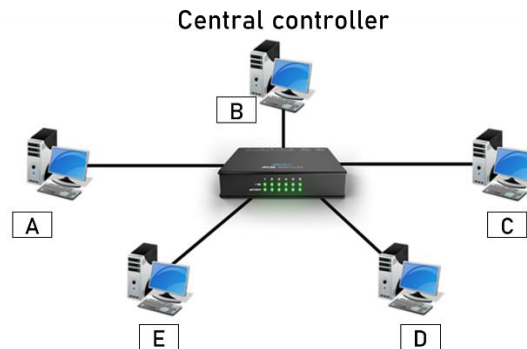


Figure 38 Hub

If A wants to send data to C, instead of sending the data directly to C, A can send the data to hub. As a result, the hub serves as a central controller, and no devices are attached directly to one another. Hub will still broadcast data, which means it will send data to C, E, and D. As a result, the hub will still broadcast data, which means it will send information to all connected devices. Following that, even the C's acknowledgment will be sent to all of the units. To begin, A sends data to C, which will arrive at the hub, which will broadcast the data, and C will accept it. Other nodes B, E, and D will search the header information and determine that the packet is not for them, so they will discard it. Then C will send the acknowledgment, and since C cannot send acknowledgements directly to A, C will send to the hub again, and the hub will broadcast the data again because this is a hub property. So it will give to B, E, and D, as well as A, with A accepting and the rest of the devices discarding the packet. To link topology segments of land and track network traffic, a hub with multiple ports is used. It organises and monitors all data sent and received between computers. Now, in order to prevent data collisions, we use the CSMA CD protocols, so if A sends data and C sends data at the same time, the data will be lost. When hub was broadcasting and C was sending the message. So, when A has already sent data to C and C has begun sending data, when the hub broadcasts the data, there will be a collision. So, to prevent these collisions, we use a CSMA CD technique, which you will understand and which will give you all of the details of this lecture when it comes up in the future lectures.

Types of HUB

Active hubs, passive hubs, and intelligent hubs are the three types of hubs as show in Figure 39. In the case of an active hub, it performs the same functions as a passive hub, but before forwarding the data signal, it amplifies it. As a result of this added functionality, active hub is also known as a repeater, as it amplifies the signal.

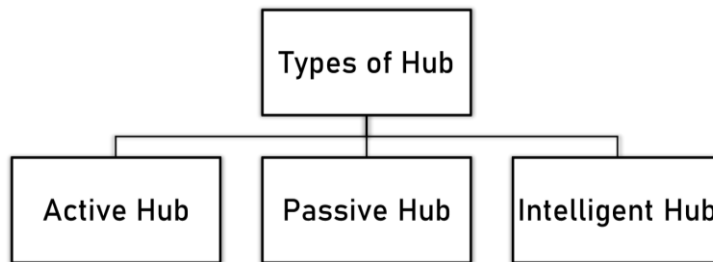


Figure 39 Types of HUB

The passive hub comes next. It transmits data signals in the same format in which they are received. It has no impact on a signal in any way. It operates in the same way as an active hub, but it also has remote control capabilities. They also offer networking devices the ability to use different data rates. It also allows an administrator to track traffic passing through the hub and configure each port in the hub, as well as determine which layer hub is active. As a result, the OSI physical layer and the TCP IP model are used by the hub.

Switch

The switch is a multiple LAN connecting system that receives data packets from multiple input ports and forwards them to a particular output port. When opposed to a hub, a switch is a more intelligent system. This is what a hub does: it takes data and sends it to all of the ports that are connected to the hub, while a switch, which is an intelligent system, receives data from one of the devices but only sends it to the specific intended recipient to whom it belongs. Let's say A wants to give some information to B. As a result, A will send data to switch, and switch will send data to C exclusively. It will not broadcast the data, but the way they are transmitting it, a switch will actually learn the physical addresses of the devices that are connected to it and store these physical addresses in a table called a MAC address.

SWITCH TABLE		
PORT	DEVICE	MAC ADDRESS
1	DETECTED	00-04-5A-63-A1-66
2	DETECTED	90-02-7B-C2-C0-67
3	DETECTED	32-07-9A-A2-00
4	DETECTED	72-00-FA-63-A9-66

Physical
Addresses

Figure 40 Switch Table

We're using a switch, which has a MAC address, which is the system's physical address. That is why we call switches intelligent devices because of how they send data to a specific recipient to which it belongs, using a MAC address to send data only to the computer to which your data belongs. But these are your MAC addresses, aren't they? 00,04,5A,63,A1,66. This is your MAC address, which is 12 digits long.

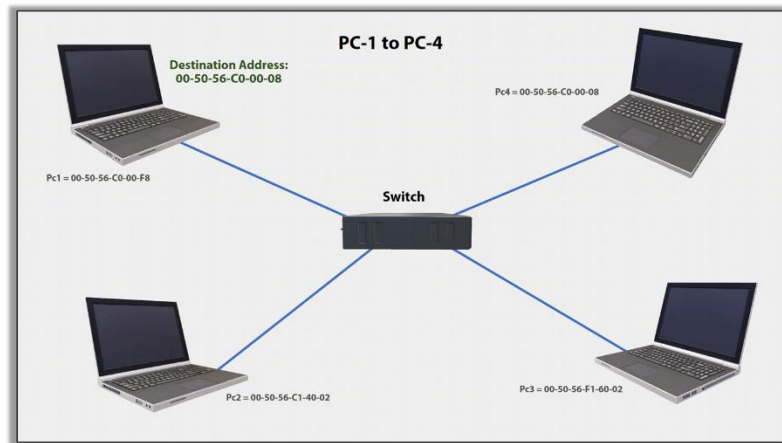


Figure 41 Switch Example

As you can see Figure 41, PC1 needs to transmit data to PC4, so PC1 sends the data to the switch, but the switch does not broadcast the data like the hub did; instead, the switch can only send the data to the intended receiver, which is PC4, since the MAC address is written on the header. As you can see, PC1 needs to transmit data to PC4, so PC1 sends the data to the switch, but the switch does not broadcast the data like the hub did; instead, the switch can only send the data to the intended receiver, which is PC4, since the MAC address is written on the header.

As a result, the MAC address is included in the post, and the data is sent to PC 4 via the MAC address switch. Only the intended recipient will receive it. As a result, the switch is aware of the MAC addresses of all systems connected to it. Now, more than ever, switch are favored over hubs because they eliminate unwanted traffic on the network.

Various Switching Methods

In the Figure 42 Different methods of switching below, various switching strategies are depicted. The first is the cut-through technique. The next choice is to use the store and forward process.

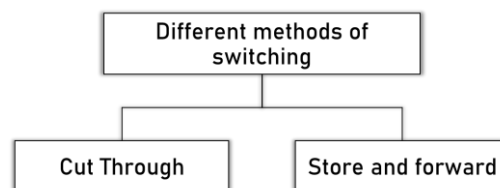


Figure 42 Different methods of switching

Cut Through

If I use the cut through process, the data will be split into small packets whenever I send it. In the case of the cut through process, the data will now be forwarded by the switch. The switch forwards the packet as soon as it arrives. As a result, each packet will take a different route to reach its destination. It does not wait for all of the data to arrive in a cut through. It simply forwards the packet as soon as it arrives. As a result, packet loss is more likely in the case of cut through and packet loss.

Store and forward

First and foremost, the switch waits for all of the data to arrive at the destination successfully before forwarding the packet to the intended recipient, then it will wait for all of the data to arrive before forwarding the packet to the destination to the intended recipient, so it will wait for all of the data to arrive before sending the packet to the destination. Packet failure is extremely unlikely in the case of store and forward strategies because it waits for all of the data to arrive before moving on to the next.

Difference between Hub and switch

Hub	Switch
Only detects that a device is physically connected to it.	Can detect specific devices that are connected to it because it keeps a record of MAC ADDRESS of those devices.

Similarity Between Hub and Switch

- Both are used to share data within a local area network, such as a home or business network.
- Both are not accustomed to exchanging data with each other outside of their own network, such as over the internet.
- Since a computer must be able to read IP addresses in order to exchange data or route data outside their own network or to another network, such as the internet.
- switch and hub Both may not read the IP address; we need a router to read the IP address. As a result, data is routed from one network to another based on IP addresses.

Router

When data packets are sent, the router inspects the IP address to decide whether the packet was intended for its own network or for another network. The router is particularly the gateway offer network. Example of router is shown in Figure 43.

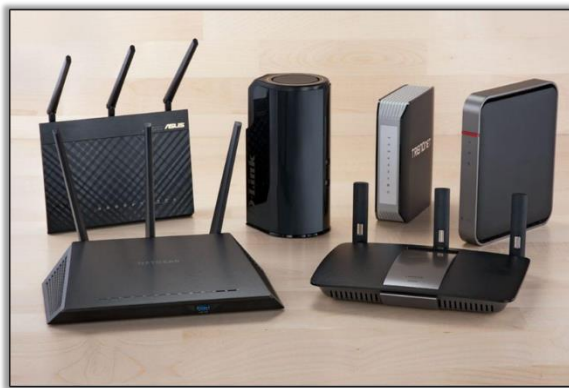


Figure 43 Router

It will allocate to the same receiver if it fits within the same network. If it isn't intended for the same network, don't use it. It is intended for use in another network, and it will transmit data to that network using IP addresses, as routers do. Routers bind two or more topologically related or dissimilar networks, such as LANs and WLANs. It will share the available bandwidth through a network of computers. As a hardware firewall, it offers better security against hacking. Routers are smart enough to figure out the shortest and quickest path from source to destination, which is the easiest way to send data from one location to another.

So, if you want to send data from source to destination, routers can assist you in finding the shortest path. There are a variety of routes to choose from. So, which path is the fastest, and which one is the shortest? Such that my packet arrives at its destination in the shortest time possible. As an example, the shortest route from node 11, N11, to node N3 is found using the router as shown in Figure 44. When a data packet is forwarded to its final destination, the linked routers are registered.

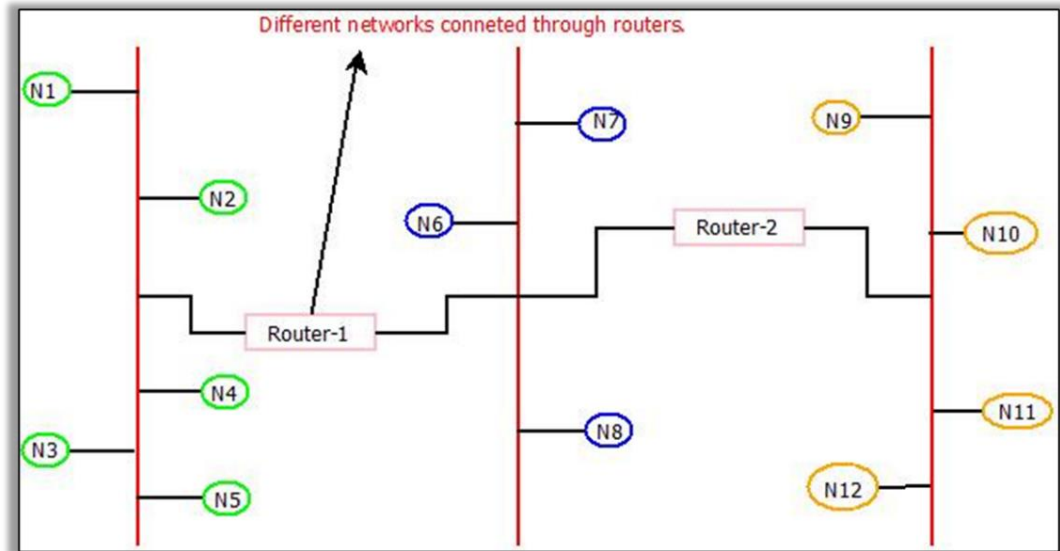


Figure 44 Description router

As a result, these records are held in a database table called the routing table. So, a routing table can be designed statistically or dynamically, so the router knows which route to take to send the data since it is updated on a regular basis. If any new networks access the network router, the destination addresses are still modified. As a result, routers are aware of each system's IP address. The IP address is the unique identifier for each device. So, routers know what the next hop is, where data will be transferred, and how long it will take to get data from source to destination. One of the functions of the router is to link many small networks into a larger network. By linking small networks together, we can create a larger network. So, one of the most popular routers, and one that is very familiar nowadays, is a wireless router, which allows a user to connect easily without installing any cables, and they also allow a user to connect easily without installing any cables, so if we are talking about wireless implies that we are not using any cables and can connect wirelessly.

Important points

- **Which Layer Router Operates?**

Operates at **network layer** of OSI model.

- When it comes to the network layer, two things must come to mind: the IP address and the router.
- the router read the system's IP address.
- The router can use this information to determine the best path for your packets or data to travel from source to destination in the shortest time possible.

Network interface card.

It is a circuit board or a card that allows computers to communicate over a network through cables or wirelessly through a small antenna network Figure 45. It connects to a network via cables cat5, coaxial cable, fibre optics, etc. When someone tells you the other term for a network interface card, it's a LAN adapter, network adapter, or network card. You may also use the terms LAN adapter, network adapter, or network card to describe your device.

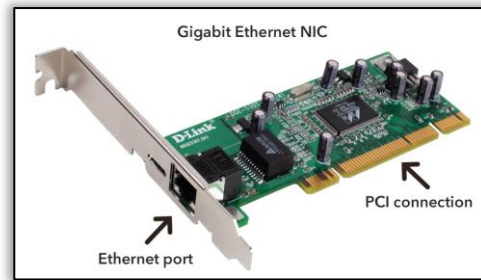


Figure 45 Network Interface Card

Types of NIC

The wired network interface card and the wireless network interface card are the two types of network interface cards.

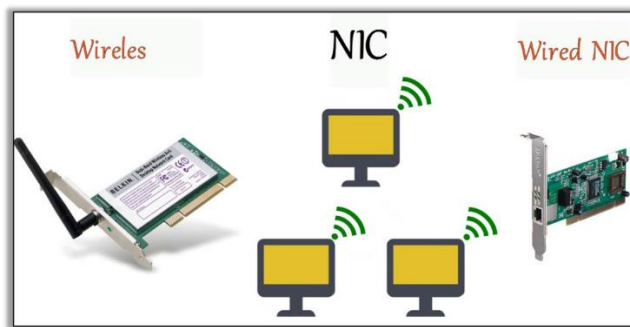


Figure 46 Types of NIC

Wireless NIC

Wireless NIC means in laptop, we are using a NIC card, it can be connected through cable. Also, but if it is a wireless NIC it can connect through the wireless modem. It works on the radio frequency.

Wired NIC

A wired network is one of the most popular wired configurations. Ethernet cables are used to transmit data between connected PCs in most wired networks. A single router will link all the computers in a small wired network. Multiple routers or switches connected to each other are common in larger networks. A cable modem, T1 line, or other form of Internet connection is usually connected to one of these devices. Wired may also apply to peripheral devices. Since several keyboards and mice are now wireless, the term "wired" is frequently used to refer to input devices that attach to a USB port as seen in Figure 47. Monitors and external hard drives use cables as well, although they are rarely referred to as wired devices because wireless alternatives are rarely available.

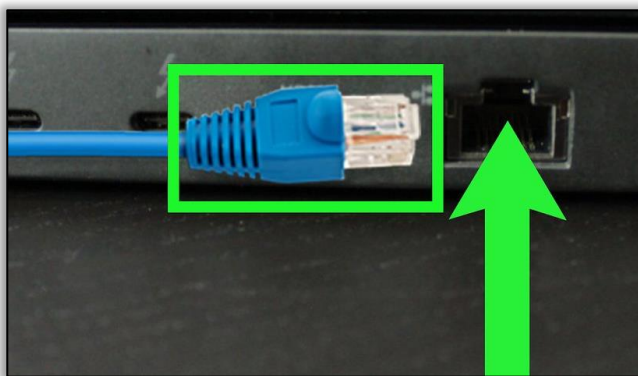


Figure 47 Wired NIC

- On desktop computer, we use wired NIC. If NIC is wired then connect through RJ45 port. By connection cable with RJ45 you communicate with internet.

How to check whether NIC card is working or not?

It's a straightforward procedure.

The first move is to go to the start menu, select Run, and type CMD into the command prompt that appears in Figure 48.

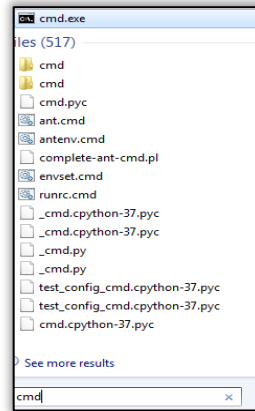


Figure 48 Open command prompt

In that case, when the company manufactures the NIC, it assigns the NIC, which is a network interface card, a loopback address in phase two. I'd like to point out that we're writing PING space 127.0.0.1 as seen in Figure 49.

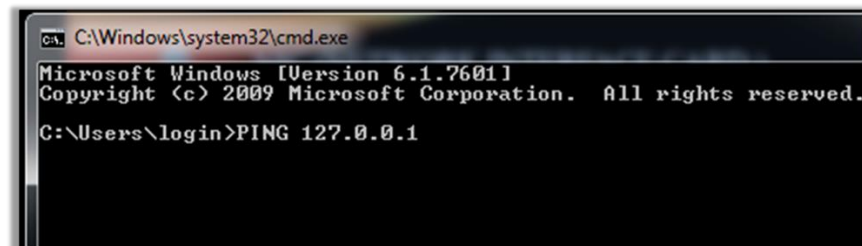


Figure 49 To check loopback address

```
C:\Users\Shivang Sri> ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 50 Reply of loopback address

So if you get a response, it means everything is fine as seen in Figure 50.

LAN Standards

Different LAN standards are:

- ISO
- ANSI
- EIA/TIA
- IEEE

But there are several LAN standards that are developing the standards that we are discussing on the NIC that we are publishing, the first of which is the ISO. So this form of organisation describes flow management, promotes flow management, and checks and promotes the consistency of the norm. The ANSI American National Standard international organisation is the next LAN standard. This is the organisation that governs all cards relevant to NIC network equipments, regulations, and quality control. The next one is your EIA, TIA, which is used when we do wired communication via cable and link one device to another via cable. Then there are two colour norms that we use. So EIA and TIA agreed on colour coding, for example, if I wanted to connect two identical devices, such as one PC to the other PC, we'd use a crossover cable. If I want to communicate, I'll use a PC with a router, which means those are two different devices. In that case, the straight cable patch straight cable would be used. So, when do you use a straight cable and when should you use a crossover cable? These are the criteria that EIA TIA decides on. The IEEE is the next organisation, which develops all communication standards. The 802.11 standard was the first land standard. 802.1, 802.2, 802.3 45678, and so on. It is essentially a correspondence that is decided by 12 bits. So latest one is 802.17 is there.

So now, a MAC address for the network interface card that we're discussing is divided into two bits, one of which is known as the manufacturing ID and the other as the serial number as seen in Figure 51.

00-04-AC	F3-1C – D4
Manufacture ID	Serial number

Figure 51 Parts of MAC Address

When a hacker tries to steal data from a target PC, the MAC address is punched on the network interface card, just like students. ISP stands for internet service provider, or security department. Cybercrime can be used to track down the intruder, and the serial number can be used to pinpoint their location. At this time, the individual is present at which site. As a result, we are able to track down the hacker using his MAC address.

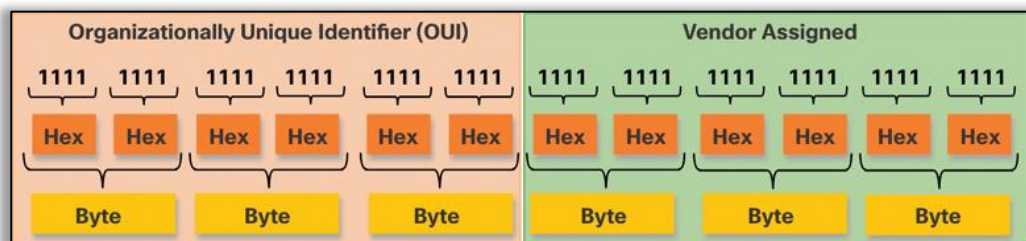


Figure 52 MAC Address format

A MAC address is a hexadecimal number of 12 digits. As a result, when you convert this to binary. As a result, it has been reduced to 48 bits. As a result, the MAC address is a long-term address. A physical address is another term for it. It is difficult to change this permanent address in practice; however, you can change the MAC address virtually with the aid of a variety of applications. Since a single hexadecimal digit represents four binary bits, an Ethernet MAC address consists of a 48-bit binary value. Hexadecimal is used to label an Ethernet address. As seen in this Figure 52 As a result, 12 hexadecimal values can be used to represent a 48-bit Ethernet MAC address.

Which Layer NIC Operates?

Operates on physical and data link layer of OSI model.

Repeater

The term "repeater" refers to a device that increases the signal's intensity as it passes from source to destination. So it's likely that your signals begin to lose energy; in fact, they begin to lose energy. As a result, we'll need an amplifier or other repeaters to increase the signal's intensity or improve it.

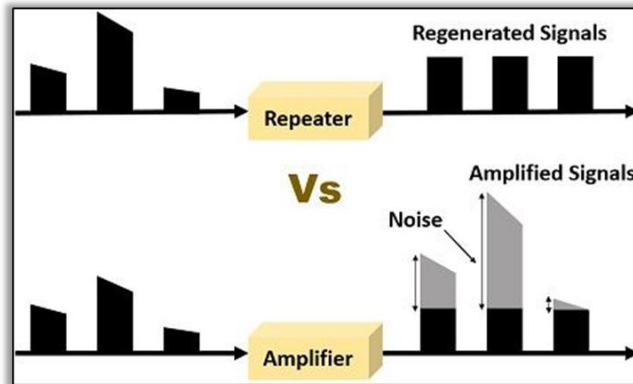


Figure 53 Repeater

It operates at the physical layer of the OSI model, and it's ideal for long-distance networks. The bus topology's main benefit is that it removes unnecessary noise from incoming signals, which is why my signal was losing energy. If you wish to receive the exact signal that is being sent, you must also give the exact signal. As a result, we'll have to use the repeater repeater.

Bridge

A bridge is a networking system that links two or more LANs together, treating them as a single network as seen in Figure 54. So, this is what distinguishes it from a router; in a router, we link two different networks together and they act as a separate network.

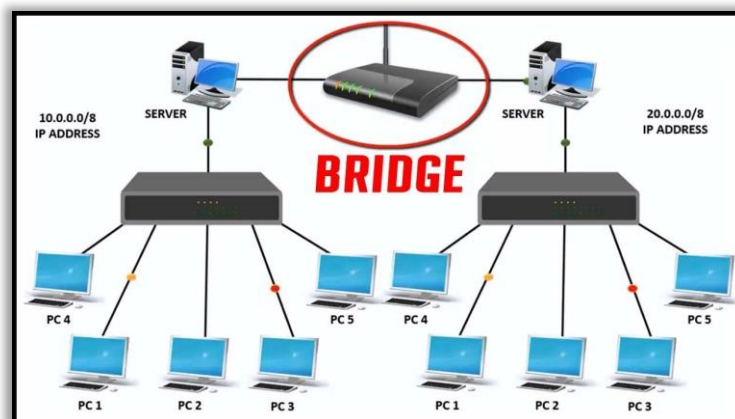


Figure 54 Bridge

A bridge is a networking system that links two or more LANs together, treating them as a single network. So this is what distinguishes it from a router; in a router, we link two different networks together and they act as a separate network. Multiple networks are linked together in this situation, but they are considered as a single network. As a result, bridges are used when the number of LANs grows and network traffic becomes too much to handle. So we use the bridge to divide the LAN network traffic into segments and thus reduce the network traffic. As a result, a bridge will send data over two separate protocols, such as Ethernet and Token Bus. It examines the frame's MAC address. If they fit, the data is forwarded; if they don't, the frame is discarded.

At which layer bridge operates?

The OSI model's data link layer is where bridge operates.

Gateway

A gateway is a computer network node (router) that serves as a critical stopover point for data on its way to or from other networks. We can connect and send data back and forth thanks to gateways. Without gateways, the Internet will be useless to us (as well as a lot of other hardware and software). The gateway is the device in an office that routes traffic from a workstation to the outside network that serves Web pages. The Internet Service Provider, which gives you access to the entire Internet, is the hub for basic Internet connections at home.

A node is essentially a physical location where data is halted for transport or reading/use. (A node is a device or modem; a computer cable is not.) A few node notes are as follows:

- A gateway or a host node may serve as a stopping point on the Internet.
- A node is a machine that manages the traffic that your Internet Service Provider (ISP) receives.

Your gateway is the modem (or modem-router combo) your ISP offers so you can connect to their network if you have a wireless network at home that allows your entire family access to the Internet. On the other hand, the device that your Internet Service Provider (ISP) uses to monitor all of the data traffic it receives and sends out is a node. When a computer server serves as a portal, it also acts as a firewall and proxy server. Unwanted traffic and outsiders are kept out of a private network by a firewall. A proxy server is software that sits between the applications you use on your computer (such as a Web browser) and the computer server that supports your network. The proxy server's job is to ensure that the real server can handle the request.

Keywords

Bandwidth: Refers to the range of frequencies assigned to a channel.

Bounded Media: Refers to the wired transmission systems that employ physical media, which are tangible.

Coaxial Cable: It is a very robust shielded copper wire two-conductor cable in which a solid center conductor runs concentrically (coaxial) inside a solid outer circular conductor.

Frequency Spectrum: Refers to the range of frequencies being supported by a particular transmission medium.

Gauge: Gauge is a measure of the thickness of the conductor.

Graded Index Multimode Fiber: In the case of a graded index multimode fiber, the index of refraction across the core is gradually changed from a maximum at the center to a minimum near the edges, hence the name graded index.

Monomode/Singlemode fiber: This has a thinner inner core. In this case, the core diameter of about 9 μm is much closer in size to the wavelength of light being propagated, about 1.3 μm . This limits the light transmission to a single ray or mode of light to propagate down the core of the fiber.

Multimode Fiber: The core diameter is relatively large compared to a wavelength of light.

Optical Fiber: Optical fiber carries the transmitted information in the form of a fluctuating beam of light in a glass fiber rather than as an electrical signal on a wire.

Cable Modem: It works on the principle of modems and provides access to data signal sent through the cable television infrastructure.

Cell Site: A circular geographical area that handles cellular phones within its defined physical boundary.

Microwave Radio: It is a form of radio transmission that uses ultra-high frequencies.

Radio: A technique where data is transmitted using radio waves and therefore energy travels through the air rather than copper or glass.

Review Questions

1. Describe how satellite communication is different from radio broadcast?
2. Explain different types of networking devices along with advantages and disadvantages.
3. What are the different transmission mediums over which data communication devices can provide service?
4. What are the major limitations of twisted pair wire?
5. Describe how satellite communication is different from radio broadcast?

Self Assessment

1. Which is the smallest unit amongst the following with reference to the ATM-
 - a) transmission path
 - b) virtual path
 - c) virtual circuit
 - d) all are of the same size
2. A device that provides a central connection point for cables is –
 - a) Switch
 - b) Hub
 - c) Gateway
 - d) Proxy Server
3. A device that helps prevent congestion and data collisions –
 - a) Switch
 - b) Hub
 - c) Gateway
 - d) Proxy Server
4. Transmission media directly controlled by
 - a) Physical layer
 - b) Data link Layer
 - c) Network Layer
 - d) Session Layer
- 5) guided media provides a conduit from one device to another, includes
 - a) twisted pair cable
 - b) fiber optic cable
 - c) coaxial cable
 - d) All of above
- 6) RG-59 is used in
 - a) radio
 - b) thick ethernet
 - c) thin ethernet
 - d) cable tv

- 7) A local telephone network is an example of a _____ network.
- a) Packet switched
 - b) Circuit switched
 - c) Bit switched
 - d) Line switched
- 8) Most packet switches use this principle _____
- a) Stop and wait
 - b) Store and forward
 - c) Store and wait
 - d) Stop and forward
- 9) In _____ systems, resources are allocated on demand.
- a) packet switching
 - b) circuit switching
 - c) line switching
 - d) frequency switching

Answers

- | | | |
|-------|-------|-------|
| 1 (c) | 2 (c) | 3 (a) |
| 4 (a) | 5 (d) | 6 (d) |
| 7 (b) | 8 (b) | 9 (a) |

Further Readings



Andrew S. Tanenbaum, Computer Networks, Prentice Hall
Behrouz A. Forouzan and Sophia Chung Fegan, Data Communications and Networking, McGraw-Hill Companies
Burton, Bill, Remote Access for Cisco Networks, McGraw-Hill Osborne Media
McGraw-Hill Osborne Media
Rajneesh Agrawal and Bharat Bhushan Tiwari, Computer Networks and Internet, Vikas Publication